

A Technical Introduction to ResNet deep learning

$$y = F(x, \{W_i\}) + x$$

Section 1

A common evaluation set for image classification is the MNIST database data set. MNIST is composed of handwritten digits and includes 60,000 training examples and 10,000 test examples. As with TIMIT, its small size lets users test multiple configurations. A comprehensive list of results on this set is available. Deep learning-based image recognition has become "superhuman", producing more accurate results than human contestants. This first occurred in 2011 in recognition of traffic signs, and in 2014, with recognition of human faces. Deep learning-trained vehicles now interpret 360° camera views. Another example is Facial Dysmorphology Novel Analysis (FDNA) used to analyze cases of human malformation connected to a large database of genetic syndromes.

Section 2

Weather prediction Traditional weather prediction systems solve a very complex system of partial differential equations. GraphCast is a deep learning based model, trained on a long history of weather data to predict how weather patterns change over time. It is able to predict weather conditions for up to 10 days globally, at a very detailed level, and in under a minute, with precision similar to state of the art systems.

Section 3

Data collection ethics The deep learning systems that are trained using supervised learning often rely on data that is created or annotated by humans, or both. It has been argued that not only low-paid clickwork (such as on Amazon Mechanical Turk) is regularly deployed for this purpose, but also implicit forms of human microwork that are often not recognized as such. The philosopher Rainer Mùhlhoff distinguishes five types of "machinic capture" of human microwork to generate training data: (1) gamification (the embedding of annotation or computation tasks in the flow of a game), (2) "trapping and tracking" (e.g. CAPTCHAs for image recognition or click-tracking on Google search results pages), (3) exploitation of social motivations (e.g. tagging faces on Facebook to obtain labeled facial images), (4) information mining (e.g. by leveraging quantified-self devices such as activity trackers) and (5) clickwork.

Section 4

The deep learning revolution started around CNN- and GPU-based computer vision. Although CNNs trained by backpropagation had been around for decades and GPU implementations of NNs for years, including CNNs, faster implementations of CNNs on GPUs were needed to progress on computer vision. Later, as deep learning becomes widespread, specialized hardware and algorithm optimizations were developed specifically for deep learning. A key advance for the deep learning revolution was hardware advances, especially GPU. Some early work dated back to 2004. In 2009, Raina, Madhavan, and Andrew Ng reported a 100M deep belief network trained on 30 Nvidia GeForce GTX 280 GPUs, an early demonstration of GPU-based deep learning. They reported up to 70 times faster training. In 2011, a CNN named DanNet by Dan Ciresan, Ueli Meier, Jonathan Masci, Luca Maria Gambardella, and Jürgen Schmidhuber achieved for the first time superhuman performance in a visual pattern recognition contest, outperforming traditional methods by a factor of 3. It then won more contests. They also showed how max-pooling CNNs on GPU improved performance significantly. In 2012, Andrew Ng and Jeff Dean created an FNN that learned to recognize higher-level concepts, such as cats, only from watching unlabeled images taken from YouTube videos. In October 2012, AlexNet by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton won the large-scale ImageNet competition by a significant margin over shallow machine learning methods. Further incremental improvements included the VGG-16 network by Karen Simonyan and Andrew Zisserman and Google's Inceptionv3. The success in image classification was then extended to the more challenging task of generating descriptions (captions) for images, often as a combination of CNNs and LSTMs. In 2014, the state of the art was training "very deep neural network" with 20 to 30 layers. Stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In 2015, two techniques were developed to train very deep networks: the highway network was published in May 2015, and the residual neural network (ResNet) in Dec 2015. ResNet behaves like an open-gated Highway Net. Around the same time, deep learning started impacting the field of art. Early examples included Google DeepDream (2015), and neural style transfer (2015), both of which were based on pretrained image classification neural networks, such as VGG-19. Generative adversarial network (GAN) by (Ian Goodfellow et al., 2014) (based on Jürgen Schmidhuber's principle of artificial curiosity) became state of the art in generative modeling during 2014-2018 period. Excellent image quality is achieved by Nvidia's StyleGAN (2018) based on the Progressive GAN by Tero Karras et al. Here the GAN generator is grown from small to large scale in a pyramidal fashion. Image generation by GAN reached popular success, and provoked discussions concerning deepfakes. Diffusion models (2015) eclipsed GANs in generative modeling since then, with systems such as

DALL·E 2 (2022) and Stable Diffusion (2022). In 2015, Google's speech recognition improved by 49% by an LSTM-based model, which they made available through Google Voice Search on smartphone. Deep learning is part of state-of-the-art systems in various disciplines, particularly computer vision and automatic speech recognition (ASR). Results on commonly used evaluation sets such as TIMIT (ASR) and MNIST (image classification), as well as a range of large-vocabulary speech recognition tasks have steadily improved. Convolutional neural networks were superseded for ASR by LSTM, but are more successful in computer vision. Yoshua Bengio, Geoffrey Hinton and Yann LeCun were awarded the 2018 Turing Award for "conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing".

Section 5

Deep Neural Network Estimations Deep neural networks can be used to estimate the entropy of a stochastic process through an arrangement called a Neural Joint Entropy Estimator (NJEE). Such an estimation provides insights on the effects of input random variables on an independent random variable. Practically, the DNN is trained as a classifier that maps an input vector or matrix X to an output probability distribution over the possible classes of random variable Y , given input X . For example, in image classification tasks, the NJEE maps a vector of pixels' color values to probabilities over possible image classes. In practice, the probability distribution of Y is obtained by a Softmax layer with number of nodes that is equal to the alphabet size of Y . NJEE uses continuously differentiable activation functions, such that the conditions for the universal approximation theorem holds. It is shown that this method provides a strongly consistent estimator and outperforms other methods in cases of large alphabet sizes.